AMIT BHUNIA

Remote Sensing & GIS Analyst

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PROFESSIONAL SUMMARY

Remote Sensing and GIS Analyst with about 2 years of experience, currently a Project Fellow at Haryana Space Applications Centre (HARSAC), Government of Haryana. Specialises in PSInSAR-based ground deformation analysis (state-wide Sentinel-1 LOS velocity mapping for Haryana) and UAV/stereo photogrammetry for DEM, DSM and orthophoto generation over mining sites using Trimble Inpho UAS Master, supported by DGPS field surveys. Also works on cadastral-level crop mapping and rice yield estimation. M.Sc. in Remote Sensing & GIS (CGPA 9.02), with NASA training in SAR and NISAR applications.

TECHNICAL SKILLS

- **GIS & Remote Sensing:** ArcGIS Pro, ArcMap, QGIS, ERDAS Imagine, ENVI, SNAP, Google Earth Engine, Global Mapper, Google Earth Pro
- **Photogrammetry:** Trimble Inpho UAS Master - UAV and stereo image processing, DEM, DSM and orthophoto generation
- **SAR / InSAR:** PSInSAR (StaMPS on MATLAB), LOS velocity mapping, Sentinel-1 time-series analysis
- **Data:** Sentinel-1, Sentinel-2, Landsat, UAV imagery, stereo imagery, hyperspectral data
- **Agriculture:** Time-series crop mapping, cadastral-level crop inventory, semi-physical rice yield modelling
- **Field Instruments:** DGPS, LAI meter, chlorophyll meter
- **Programming & CAD:** MATLAB (StaMPS), Python (basic), R (basic), AutoCAD
- **AI & Automation:** n8n workflow automation, local open-source LLMs (Qwen) via LM Studio API, RAG

PROFESSIONAL EXPERIENCE

Project Fellow | Haryana Space Applications Centre (HARSAC), Government of Haryana | Dec 2024 - Present

- Derive InSAR line-of-sight (LOS) velocity products from Sentinel-1 SAR data to map ground deformation across the entire state of Haryana, used for groundwater verification and landslide prediction.
- Generate DEMs, DSMs, DTMs and orthophotos for 7+ mining sites across Haryana from stereo and UAV imagery using Trimble Inpho UAS Master, applied to illegal mining detection and excavation volume estimation.
- Perform satellite time-series crop mapping down to cadastral (parcel) level and estimate rice yield using a semi-physical model.
- Process and analyse multispectral, hyperspectral and SAR satellite imagery in ArcGIS Pro and SNAP for research and project development.
- Collect field and ground-truth data using DGPS, LAI meter and chlorophyll meter to support satellite and photogrammetry products.

Junior GIS Engineer | NeoGeoInfo Technologies Pvt. Ltd. | Sep 2024 - Dec 2024

- Produced 50+ maps for client GIS projects using ArcGIS and Global Mapper.
- Created and maintained geospatial datasets, integrating AutoCAD drawings into GIS workflows.

PROJECTS

Land Subsidence Monitoring in Kolkata using Sentinel-1 PSInSAR

- Applied Persistent Scatterer InSAR (PSInSAR) to 21 Sentinel-1 images spanning 2019-2024, processed in StaMPS (MATLAB), to measure ground displacement across Kolkata.
- Mapped LOS velocities from -30 to +20 mm/year, identifying subsidence-prone zones relevant to urban infrastructure planning.

Landslide Susceptibility Zonation of West and South Sikkim using AHP

- Built an Analytical Hierarchy Process (AHP) weighted multi-criteria model in a GIS environment to assess landslide susceptibility.
- Produced landslide susceptibility zone maps with recommendations for disaster management and land-use planning.

AI-Driven LinkedIn Post Automation (n8n + Local LLM + Telegram)

- Built an n8n workflow that takes natural-language instructions via a Telegram bot and drafts LinkedIn posts using the open-source Qwen LLM served locally through the LM Studio API.
- Added a human-in-the-loop approval step in Telegram so each draft is reviewed before auto-publishing to LinkedIn, with no paid cloud LLM API.

EDUCATION

M.Sc. in Remote Sensing & GIS | Vidyasagar University, Midnapore, West Bengal | 2022 - 2024 | CGPA: 9.02

B.Sc. in Geography | Khejuri College, Purba Medinipur, West Bengal | 2019 - 2022 | CGPA: 9.07

CERTIFICATIONS

- An Introduction to Synthetic Aperture Radar (SAR) and Its Applications - NASA
- Harnessing NISAR: Next-Generation Radar Observations for Earth Applications - NASA

LANGUAGES

Bengali (native), Hindi, English